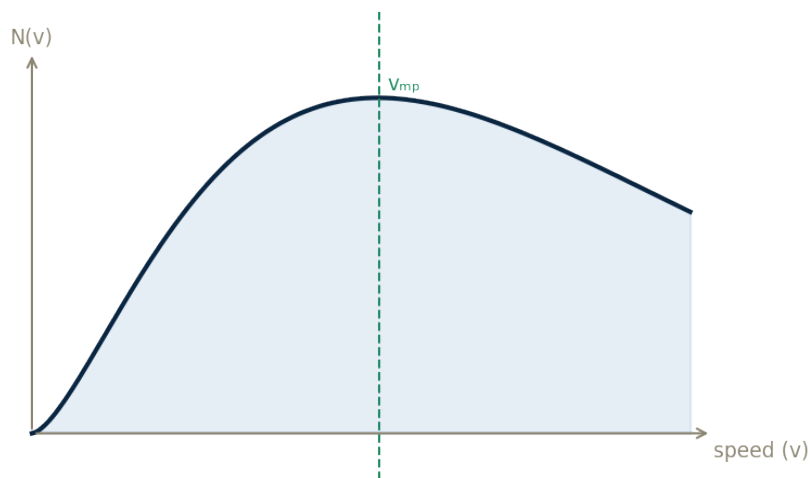


Kinetic Theory of Gases

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Kinetic theory models a gas as a large number of point particles in constant random motion, colliding elastically with each other and the container walls.
- Ideal gas equation: $PV = nRT$, combining Boyle's, Charles's, and Avogadro's laws.
- Pressure from kinetic theory: $P = \frac{1}{3} \rho (v_{\text{rms}})^2$, where ρ is density and v_{rms} is the root-mean-square speed.
- Average kinetic energy per molecule = $\frac{3}{2} k_B T$ (k_B = Boltzmann constant), independent of the gas's nature - this is the basis of the equipartition principle.
- Degrees of freedom: monatomic gas = 3 (translational only); diatomic gas = 5 (3 translational + 2 rotational) at moderate temperatures.
- Law of equipartition of energy: each degree of freedom contributes $\frac{1}{2} k_B T$ of energy per molecule on average.
- Mean free path (λ) is the average distance a molecule travels between successive collisions; λ is inversely proportional to the number density and the molecular cross-section.



Maxwell-Boltzmann speed distribution curve showing the most probable speed $v(mp)$.

2. Formula Sheet

Ideal gas equation	$PV = nRT$
Pressure from kinetic theory	$P = \frac{1}{3} \rho v_{rms}^2$
RMS speed	$v_{rms} = \sqrt{3RT/M} = \sqrt{3k_B T/m}$
Average KE per molecule	$KE(\text{avg}) = \frac{3}{2} k_B T$
Average KE per mole	$U = \frac{3}{2} RT$ (monatomic)
Degrees of freedom energy	$E = \frac{f}{2} k_B T$ per molecule
Most probable speed	$v_{mp} = \sqrt{2RT/M}$
Average speed	$v_{avg} = \sqrt{8RT/(\pi M)}$

3. Practice Problems (22 questions)

- Find the RMS speed of oxygen molecules ($M = 32 \text{ g/mol}$) at 300 K ($R = 8.31 \text{ J/mol K}$).
- One mole of an ideal gas occupies 22.4 L at STP ($T = 273 \text{ K}$, $P = 1 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$). Verify this using $PV = nRT$.
- Find the average kinetic energy of a gas molecule at 27 degrees C ($k_B = 1.38 \times 10^{-23} \text{ J/K}$).
- A gas is heated so that its absolute temperature doubles. Find the factor by which the RMS speed of its molecules changes.
- State the assumptions of the kinetic theory of gases regarding molecular collisions.
- For a diatomic gas, find the total internal energy of 2 moles at temperature T using the equipartition theorem ($5 \text{ degrees of freedom}$).
- Compare the RMS speeds of hydrogen ($M = 2 \text{ g/mol}$) and oxygen ($M = 32 \text{ g/mol}$) molecules at the same temperature. Find the ratio $v_{rms}(H_2) : v_{rms}(O_2)$.
- Explain why the pressure of a gas increases when its temperature is increased at constant volume, using the kinetic theory.
- Find the RMS speed of nitrogen molecules ($M = 28 \text{ g/mol}$) at 0 degrees C .
- State the law of equipartition of energy and use it to find C_v for a monatomic ideal gas.
- Two gases, helium and argon, are at the same temperature. Which gas has molecules with a higher average speed, and why?
- The pressure of a fixed mass of gas is doubled at constant temperature. Find how the density of the gas changes.
- Define mean free path and state how it depends on the number density of gas molecules.
- Find the number of moles in 4 L of an ideal gas at 2 atm and 300 K ($R = 0.0821 \text{ L atm/mol K}$).
- Explain, using kinetic theory, why gases are more compressible than liquids or solids.
- A gas at 300 K has RMS speed v . Find the temperature at which the RMS speed becomes $2v$.

17. State Avogadro's law and explain its molecular interpretation.
18. Find the total translational kinetic energy of 1 mole of an ideal gas at 400 K ($R = 8.31 \text{ J/mol K}$).
19. Why does helium gas escape from a balloon faster than oxygen, given similar-sized pores in the balloon material?
20. State the relationship between the RMS speed, average speed, and most probable speed of gas molecules (in terms of relative magnitude).
21. A closed container has gas at pressure P and absolute temperature T . If the temperature is increased to $1.5T$ at constant volume, find the new pressure.
22. Explain what is meant by an 'ideal gas' and state two ways in which real gases deviate from ideal gas behavior at high pressure or low temperature.

Answer Key

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| 1. v_{rms} approx 483 m/s | 2. Conceptual/numerical check - $PV = nRT$ holds (approx 22.4 L) |
| 3. $KE(\text{avg})$ approx $6.21 \times 10^{-21} \text{ J}$ | 4. Factor = $\sqrt{2}$ approx 1.41 |
| 5. Conceptual - elastic collisions, negligible molecular size and intermolecular forces, random motion | 6. $U = 5RT$ (2 moles $\times (5/2)RT$) |
| 7. Ratio = 4 : 1 ($\sqrt{32/2}$) | 8. Conceptual - higher T increases molecular speed, so more frequent/forceful collisions with walls, raising pressure |
| 9. v_{rms} approx 493 m/s | 10. Conceptual - each DOF contributes $(1/2)kBT$; $C_v = (3/2)R$ for monatomic gas (3 DOF) |
| 11. Helium - lower molar mass gives higher RMS/average speed at same T | 12. Density doubles (at constant T , $PV=nRT$ means density proportional to P) |
| 13. Conceptual - inversely proportional to number density and molecular cross-sectional area | 14. n approx 0.325 mol |
| 15. Conceptual - large intermolecular spacing and weak forces allow easy compression | 16. $T' = 1200 \text{ K}$ (RMS speed proportional to $\sqrt{T}$) |
| 17. Conceptual - equal volumes of gases at same T , P contain equal numbers of molecules | 18. $KE(\text{total}) = (3/2)RT \times n = 4986 \text{ J}$ (for 1 mole) |
| 19. Conceptual - lower molar mass gives higher RMS speed, so helium effuses/diffuses faster | 20. Conceptual - $v_{\text{rms}} > v_{\text{avg}} > v_{\text{mp}}$ |
| 21. $P' = 1.5 P$ | 22. Conceptual - ideal gas has point molecules with no intermolecular forces; real gases deviate due to finite molecular size and attractive forces at high P /low T |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.